The Kalman Filter

Linear Gaussian state-space models: theory, proofs and the `nabia` implementation

nabia documentation

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Abstract

This note documents `KalmanFilter`, the `nabia` forecaster for linear Gaussian state-space models with a scalar observation. We first prove the Gaussian conditioning lemma and derive the filter from it. We then prove that the Kalman gain minimizes the error covariance among all linear updates, in the Loewner order, and show why the Joseph form of the covariance update is used. We derive the likelihood by the prediction error decomposition, the h-step forecast distribution, the treatment of missing values and the Rauch–Tung–Striebel smoother. For the two built-in models, local level and local linear trend, we derive the steady state, the link with exponential smoothing and the growth of the forecast variance with the horizon. The last section maps the algorithm to the code and discusses numerical choices, complexity and limitations.

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1 Model

A linear Gaussian state-space model with n states and a scalar observation is

$$\begin{aligned} x_t &= Fx_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q), \\ y_t &= Hx_t + v_t, \qquad v_t \sim \mathcal{N}(0, R), \end{aligned} \qquad t = 0, \dots, T-1, \tag{1}$$

with transition $F \in \mathbb{R}^{n\times n}$, observation row $H \in \mathbb{R}^{1\times n}$, process noise covariance $Q \succeq 0$ and observation noise variance $R > 0$. The prior on the first state is $x_0 \sim \mathcal{N}(\bar{x}_0, P_0)$. The initial state, the w_t and the v_t are mutually independent and independent over time. The state x_t is hidden; only y_t is observed, and some y_t may be missing.

Notation. For $s \le t$, write $y_{0:s}$ for the observations up to time s and

$$x_{t|s} = \mathbb{E}[x_t \mid y_{0:s}], \qquad P_{t|s} = \mathrm{Cov}(x_t \mid y_{0:s}).$$

With this notation, $x_{0|-1} = \bar{x}_0$ and $P_{0|-1} = P_0$. The *filter* computes $(x_{t|t}, P_{t|t})$, the *one-step predictor* $(x_{t|t-1}, P_{t|t-1})$, the *forecast* $(x_{T-1+h|T-1}, P_{T-1+h|T-1})$ and the *smoother* $(x_{t|T-1}, P_{t|T-1})$.

2 The filter

2.1 Gaussian conditioning

Lemma 2.1 (Conditioning a Gaussian vector)**.** *Let (a, b) be jointly Gaussian with means μ_a, μ_b, covariances Σ_{aa}, Σ_{bb}, cross-covariance $\Sigma_{ab} = \Sigma_{ba}^\top$, and $\Sigma_{bb} \succ 0$. Then*

$$a \mid b \;\sim\; \mathcal{N}(\mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b),\; \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}).$$

Proof. Let $G = \Sigma_{ab}\Sigma_{bb}^{-1}$ and $c = a - Gb$. Then $\mathrm{Cov}(c, b) = \Sigma_{ab} - G\Sigma_{bb} = 0$. Since (c, b) is a linear function of the Gaussian vector (a, b), it is jointly Gaussian, and zero covariance implies that c and b are independent. Writing $a = c + Gb$, conditionally on b the vector a is c (whose distribution is unchanged by conditioning) plus the constant Gb. Hence $a \mid b$ is Gaussian with mean $\mathbb{E}c + Gb = \mu_a - G\mu_b + Gb$ and covariance $\mathrm{Cov}(c) = \Sigma_{aa} - G\Sigma_{ba} - \Sigma_{ab}G^\top + G\Sigma_{bb}G^\top = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$. □

2.2 Prediction and correction

Theorem 2.2 (Kalman filter)**.** *For model (1), every distribution $x_t \mid y_{0:s}$ is Gaussian, and the moments obey, for $t \ge 0$:*

$$\textit{predict } (t \ge 1)\text{:} \quad x_{t|t-1} = Fx_{t-1|t-1}, \qquad P_{t|t-1} = FP_{t-1|t-1}F^\top + Q, \tag{2}$$

$$\textit{innovation:} \quad e_t = y_t - Hx_{t|t-1}, \qquad S_t = HP_{t|t-1}H^\top + R, \tag{3}$$

$$\textit{gain:} \quad K_t = P_{t|t-1}H^\top / S_t, \tag{4}$$

$$\textit{correct:} \quad x_{t|t} = x_{t|t-1} + K_t e_t, \qquad P_{t|t} = (I - K_t H)\, P_{t|t-1}. \tag{5}$$

Moreover $y_t \mid y_{0:t-1} \sim \mathcal{N}(Hx_{t|t-1}, S_t)$.

Proof. By induction on t. Assume $x_{t-1} \mid y_{0:t-1} \sim \mathcal{N}(x_{t-1|t-1}, P_{t-1|t-1})$. For $t = 0$, the prior plays this role and there is no prediction step.

Prediction. The noise w_t is independent of $(x_{t-1}, y_{0:t-1})$. Given $y_{0:t-1}$, the state $x_t = Fx_{t-1} + w_t$ is an affine function of two independent Gaussian vectors. So $x_t \mid y_{0:t-1}$ is Gaussian with mean $Fx_{t-1|t-1}$ and covariance $FP_{t-1|t-1}F^\top + Q$.

Correction. The noise v_t is independent of $(x_t, y_{0:t-1})$, so given $y_{0:t-1}$ the pair (x_t, y_t) is jointly Gaussian with

$$\mathbb{E}[y_t \mid y_{0:t-1}] = Hx_{t|t-1}, \qquad \mathrm{Var}(y_t \mid y_{0:t-1}) = HP_{t|t-1}H^\top + R = S_t,$$

$$\mathrm{Cov}(x_t, y_t \mid y_{0:t-1}) = P_{t|t-1}H^\top.$$

This already proves the claim on $y_t \mid y_{0:t-1}$. Conditioning additionally on y_t means applying Lemma 2.1 to this conditional joint distribution, with $a = x_t$, $b = y_t$ and $\Sigma_{bb} = S_t \ge R > 0$. This gives the mean $x_{t|t-1} + P_{t|t-1}H^\top S_t^{-1} e_t$ and covariance $P_{t|t-1} - P_{t|t-1}H^\top S_t^{-1} HP_{t|t-1} = (I - K_t H)P_{t|t-1}$. □

With a scalar observation, S_t is a positive number. The gain is a vector, and no matrix is ever inverted.

2.3 Optimality of the gain and the Joseph form

Theorem 2.2 uses Gaussianity. Without it, the filter is still the best *linear* update.

Proposition 2.3 (Joseph form, valid for any gain)**.** *Let $\hat{x}^-$ be an estimate of x with error covariance $P^- = \mathrm{Cov}(x - \hat{x}^-)$, and observe $y = Hx + v$ with v zero-mean, variance R, and uncorrelated with $x - \hat{x}^-$. For any gain $G \in \mathbb{R}^n$, the update $\hat{x}^+ = \hat{x}^- + G(y - H\hat{x}^-)$ has error covariance*

$$P^+(G) = (I - GH)\,P^-\,(I - GH)^\top + GRG^\top. \tag{6}$$

Proof. The new error is $x - \hat{x}^+ = (x - \hat{x}^-) - G(Hx + v - H\hat{x}^-) = (I - GH)(x - \hat{x}^-) - Gv$. The two terms are uncorrelated, so their covariances add, giving (6). □

Theorem 2.4 (The Kalman gain is Loewner-optimal)**.** *With $S = HP^-H^\top + R$ and $K = P^-H^\top/S$, for every G,*

$$P^+(G) = (I - KH)P^- + (G - K)\,S\,(G - K)^\top \;\succeq\; P^+(K) = (I - KH)P^-.$$

In particular K minimizes $\operatorname{tr} P^+(G)$ (the mean squared error) and every other weighted mean squared error $\mathbb{E}\|x - \hat{x}^+\|_W^2$, $W \succeq 0$.

Proof. Expanding (6), $P^+(G) = P^- - GHP^- - P^-H^\top G^\top + GSG^\top$. Using $P^-H^\top = KS$ and completing the square,

$$P^+(G) = P^- - KSK^\top + (G - K)S(G - K)^\top.$$

Since $KSK^\top = KHP^-$, the first two terms equal $(I - KH)P^-$. The last term is positive semi-definite because $S > 0$, and it vanishes iff $G = K$. For the weighted error, $\mathbb{E}\|e\|_W^2 = \operatorname{tr}(WP^+)$ and $\operatorname{tr}(W(G - K)S(G - K)^\top) \geq 0$. □

Why the implementation uses the Joseph form. The short form $(I - KH)P^-$ equals (6) only for the exact gain. In floating point the computed K is slightly off, and the short form then loses symmetry and can even lose positive semi-definiteness: it subtracts two nearly equal matrices. Equation (6) is a sum of two positive semi-definite terms, so it stays symmetric positive semi-definite for any K. By Theorem 2.4, an error δ in the gain changes it only by $\delta S\delta^\top = O(\|\delta\|^2)$. `nabia` therefore updates the covariance as $P \leftarrow (I - KH)P(I - KH)^\top + R\,KK^\top$.

2.4 Missing observations

If y_t is missing, conditioning on the available data is conditioning on $y_{0:t-1}$ only. Then $x_{t|t} = x_{t|t-1}$ and $P_{t|t} = P_{t|t-1}$: the correction step is skipped. Uncertainty grows through consecutive prediction steps and shrinks again at the next observation. `nabia` treats any non-finite y_t (NaN) as missing.

3 Likelihood

Proposition 3.1 (Prediction error decomposition)**.** *Let $\mathcal{O} \subseteq \{0, \dots, T-1\}$ be the set of observed times. The log density of the observed values is*

$$\log p\big((y_t)_{t\in\mathcal{O}}\big) = -\frac{1}{2}\sum_{t\in\mathcal{O}}\Big(\log(2\pi) + \log S_t + \frac{e_t^2}{S_t}\Big).$$

Proof. By the chain rule of probability, the joint density of the observed values is the product of the conditional densities $p(y_t \mid \text{observed } y_s,\ s < t)$ over $t \in \mathcal{O}$. By Theorem 2.2, applied with the missing values skipped, each factor is $\mathcal{N}(y_t;\, Hx_{t|t-1}, S_t)$. Taking logs gives the result. □

The likelihood therefore comes for free with the filter, in $O(1)$ extra work per step. It is the objective for estimating unknown entries of F, H, Q, R by maximum likelihood.

Remark 3.2 (Diffuse prior). With a large P_0 (the `nabia` default is 10^7), the first term contains $\log S_0 \approx \log P_0$. This term reflects the vague prior, not the data. When comparing likelihoods across parameter values with the same P_0, the term is nearly constant and harmless. To compare models with different state dimensions, drop the first n terms (the diffuse likelihood).

4 Forecasting

Proposition 4.1 (h-step forecast)**.** *For $h \geq 1$, $y_{T-1+h} \mid y_{0:T-1} \sim \mathcal{N}(\mu_h, v_h)$ with*

$$x_{T-1+h|T-1} = F^h x_{T-1|T-1}, \qquad P_{T-1+h|T-1} = FP_{T-2+h|T-1}F^\top + Q,$$
$$\mu_h = Hx_{T-1+h|T-1}, \qquad v_h = HP_{T-1+h|T-1}H^\top + R.$$

Proof. Future observations are missing data. Apply the prediction step (2) h times without corrections, then take the observation moments as in the proof of Theorem 2.2. □

5 Smoothing

The filter uses only the past. For analysis (for example, extracting the trend of a historical series), the smoother uses the whole sample.

Theorem 5.1 (Rauch–Tung–Striebel smoother [2])**.** *Let $x_{T-1|T-1}, P_{T-1|T-1}$ be the last filtered moments. For $t = T-2, \ldots, 0$, with $C_t = P_{t|t}F^\top P_{t+1|t}^{-1}$,*

$$x_{t|T-1} = x_{t|t} + C_t\,(x_{t+1|T-1} - x_{t+1|t}), \qquad P_{t|T-1} = P_{t|t} + C_t\,(P_{t+1|T-1} - P_{t+1|t})\,C_t^\top.$$

Proof. Step 1 (Markov property). Given x_{t+1}, the state x_t is independent of $y_{t+1:T-1}$. Those observations are functions of x_{t+1} and of the noises $w_{t+2:T-1}, v_{t+1:T-1}$, which are independent of x_t and $y_{0:t}$. Hence $p(x_t \mid x_{t+1}, y_{0:T-1}) = p(x_t \mid x_{t+1}, y_{0:t})$.

Step 2. Given $y_{0:t}$, the pair (x_t, x_{t+1}) is jointly Gaussian with $\operatorname{Cov}(x_t, x_{t+1}) = P_{t|t}F^\top$ and $\operatorname{Cov}(x_{t+1}) = P_{t+1|t}$. Lemma 2.1 gives $x_t \mid x_{t+1}, y_{0:t} \sim \mathcal{N}(x_{t|t} + C_t(x_{t+1} - x_{t+1|t}),\ P_{t|t} - C_tP_{t+1|t}C_t^\top)$.

Step 3. Average over $x_{t+1} \mid y_{0:T-1} \sim \mathcal{N}(x_{t+1|T-1}, P_{t+1|T-1})$. The law of total expectation gives the mean. The law of total variance gives $P_{t|t} - C_tP_{t+1|t}C_t^\top + C_tP_{t+1|T-1}C_t^\top$, which is the stated covariance. □

`KalmanFilter::smoothed_states()` returns the smoothed means. It computes C_t as $\left(P_{t+1|t}^{-1}FP_{t|t}\right)^\top$ with an $LDL^\top$ solve, using the symmetry of both covariances, rather than forming an inverse.

6 Built-in models

6.1 Local level

$F = H = 1$, $Q = q$, $R = r$: a random walk observed with noise. The predicted variance follows the scalar Riccati recursion $P_{t+1|t} = P_{t|t-1}r/(P_{t|t-1} + r) + q$.

Proposition 6.1 (Steady state)**.** *The predicted variance converges to* $P = \frac{1}{2}(q + \sqrt{q^2 + 4qr})$ *and the gain to* $K = P/(P + r)$*. The filter then becomes simple exponential smoothing with* $\alpha = K$*, and* $q/r = K^2/(1 - K)$*.*

Proof. A fixed point solves $P^2 - qP - qr = 0$, whose positive root is the stated P. The map $g(P) = Pr/(P + r) + q$ is increasing with $g'(P) = r^2/(P + r)^2 < 1$ for $P > 0$, and it is concave. Hence the iterates are monotone and bounded, and they converge to the unique positive fixed point from any $P_0 \geq 0$. With $F = H = 1$, (5) reads $x_{t|t} = x_{t-1|t-1} + K_t(y_t - x_{t-1|t-1})$, the SES recursion. The relation $q/r = K^2/(1 - K)$ follows from substituting $P = Kr/(1 - K)$ into $P^2 = q(P + r)$. □

Diffuse start. With $P_0 \gg r$, the first gain is $K_0 = P_0/(P_0 + r) \approx 1$, so $x_{0|0} \approx y_0$ and $P_{0|0} = P_0 r/(P_0 + r) \approx r$. A large initial variance thus amounts to starting from the first observation, as SES does.

6.2 Local linear trend

With state $x_t = (\ell_t, b_t)$ (level and slope),

$$F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad Q = \operatorname{diag}(q_\ell, q_b).$$

Proposition 6.2 (Trend extrapolation and variance growth)**.** *The point forecast is the straight line* $\mu_h = \ell_{T-1|T-1} + h\, b_{T-1|T-1}$*. The part of the forecast variance due to future slope shocks equals* $q_b \sum_{m=1}^{h-1} m^2 = q_b\,(h-1)h(2h-1)/6 \sim q_b h^3/3$*.*

Proof. By induction, $F^h = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$, since $F^{h+1} = F^h F = \begin{pmatrix} 1 & h+1 \\ 0 & 1 \end{pmatrix}$. With Proposition 4.1, this gives the mean. For the variance, unroll the model from $T-1$: $b_{T-1+k} = b_{T-1} + \sum_{i=1}^{k} w^b_{T-1+i}$, and

$$\ell_{T-1+h} = \ell_{T-1} + \sum_{k=0}^{h-1} b_{T-1+k} + \sum_{j=1}^{h} w^\ell_{T-1+j} = \ell_{T-1} + h\, b_{T-1} + \sum_{i=1}^{h-1} (h-i)\, w^b_{T-1+i} + \sum_{j=1}^{h} w^\ell_{T-1+j}.$$

The future shocks are independent of the current state. The slope shocks contribute $q_b \sum_{i=1}^{h-1} (h - i)^2 = q_b \sum_{m=1}^{h-1} m^2$. □

So a trend model extrapolates the current slope. Its uncertainty grows like h^3, much faster than the linear growth of the local level model, because any error in the slope accumulates over the horizon. A small q_b gives a nearly deterministic trend, and $q_b = 0$ gives a fixed slope estimated from the data.

7 Implementation

7.1 Interface

```
auto p = std::make_unique<KalmanFilterParams>(
    KalmanFilterParams::local_linear_trend(/*q_level*/ 1e-4, /*q_slope*/ 1e-8, /*R
        */ 0.09));
KalmanFilter kf(std::move(p));
kf.fit(y); // NaN = missing
Forecast fc = kf.forecast(10); // mean and variance (Prop. 4.1)
const Eigen::MatrixXd& x = kf.filtered_states(); // n x T, column t = x_{t|t}
Eigen::MatrixXd xs = kf.smoothed_states(); // n x T, column t = x_{t|T-1}
double ll = kf.log_likelihood(); // Prop. 3.1
```

To define any other model, fill in the parameter fields:

`transition`	F	`observation_noise`	R
`observation`	H	`initial_state`	$\bar{x}_0$
`process_noise`	Q	`initial_covariance`	P_0

7.2 Algorithm

Algorithm 1 `KalmanFilter::fit`

$x \leftarrow \bar{x}_0$; $P \leftarrow P_0$; $\ell \leftarrow 0$
for $t = 0, \dots, T-1$ **do**
  if $t > 0$ **then** $x \leftarrow Fx$; $P \leftarrow FPF^\top + Q$
  end if ▷ predict
  store $x_{t|t-1} = x$, $P_{t|t-1} = P$
  $\hat{y}_t \leftarrow Hx$; $u \leftarrow PH^\top$; $S_t \leftarrow Hu + R$ ▷ u reused by the gain
  if y_t is finite **then**
    $e_t \leftarrow y_t - \hat{y}_t$; $K \leftarrow u/S_t$; $x \leftarrow x + Ke_t$
    $A \leftarrow I - KH$; $P \leftarrow APA^\top + RKK^\top$ ▷ Joseph form
    $\ell \leftarrow \ell - \frac{1}{2}(\log 2\pi + \log S_t + e_t^2/S_t)$
  else $e_t \leftarrow$ NaN ▷ missing: no correction
  end if
  store $x_{t|t} = x$, $P_{t|t} = P$
end for

Complexity. Each step costs $O(n^3)$ for the products with F and $O(n^2)$ for the rank-one correction, so fitting is $O(Tn^3)$. The filter stores the predicted and filtered moments ($O(Tn^2)$ memory) because the smoother needs them. Forecasting costs $O(hn^3)$, and smoothing $O(Tn^3)$. For the built-in models $n \le 2$, and the cost is linear in T.

Validation. The constructor checks that $F, H, Q, \bar{x}_0, P_0$ have consistent dimensions, that Q and P_0 are square and symmetric (to a relative tolerance of 10^{-10}), and that $R > 0$ is finite. It throws `std::invalid_argument` otherwise. `fit` rejects an empty series. Accessors called before `fit` throw `std::logic_error`.

Correspondence with the code.

Mathematics	`KalmanFilter`
$x_{t\|t}$, $P_{t\|t}$	`filtered_states().col(t)`, `filtered_covariances()[t]`
$Hx_{t\|t-1}$	`fitted_values()(t)`
e_t, S_t	`innovations()(t)`, `innovation_variances()(t)`
$\log p(y)$	`log_likelihood()`
$x_{t\|T-1}$	`smoothed_states().col(t)`
μ_h, v_h	`forecast(H).mean(h-1)`, `.variance(h-1)`

Tests. `tests/test_forecasters.cpp` checks:

- Proposition 6.1: the filter matches SES with the steady-state gain to 10^{-9}.
- Proposition 3.1: the log-likelihood equals the log density of $y \sim \mathcal{N}(\bar{x}_0 \mathbf{1}, \Sigma)$ with $\Sigma_{ij} = P_0 + q \min(i, j) + r\, \delta_{ij}$, computed directly, to 10^{-10}.

- Proposition 6.2: the slope state recovers a true slope of 0.5, and forecasts lie on the extrapolated line.
- Proposition 4.1: for the local level, $v_h = P_{T-1|T-1} + hq + r$.
- Missing values: the variance grows at missing steps and shrinks again at the next observation.
- Smoother: with $q = 0$, every smoothed state equals the final filtered estimate.

7.3 Limitations and extensions

- Observations are scalar and the system matrices are time-invariant. Vector observations would replace the division by S_t with a Cholesky solve.
- Q, R (and any unknowns in F, H) are inputs. A maximum-likelihood fit can maximize `log_likelihood()` over log-variances, for example with `NelderMead` or, using analytic score recursions, with `LBFGS`.
- The smoother returns means only. Theorem 5.1 also gives the smoothed covariances if needed.

References

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